

P. 9-16 From Eq. (9-103): $f_{mnp} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$.

$a = 0.08 \text{ (m)}, b = 0.06 \text{ (m)}, d = 0.05 \text{ (m)}.$

$f_{mnp} = 1.5 \times 10^8 F(m, n, p), F(m, n, p) = 100 \sqrt{\left(\frac{m}{8}\right)^2 + \left(\frac{n}{6}\right)^2 + \left(\frac{p}{5}\right)^2}.$

Eight lowest-order modes and their resonant frequencies:

Modes	$F(m, n, p)$	f_{mnp} in (GHz)
TM_{110}	20.83	3.125
TE_{101}	23.58	3.538
TE_{011}	26.03	3.905
TE_{111}, TM_{111}	28.88	4.332
TM_{210}	30.05	4.507
TE_{201}	32.02	4.802
TM_{120}	35.60	5.340

P. 9-17 a) Since $d > a > b$, the lowest-order resonant mode is TE_{101} mode.

$$f_{101} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{d^2}} = 4.802 \times 10^9 \text{ (Hz)} = 4.802 \text{ (GHz)}.$$

b) From Eq. (9-120):

$$Q_{101} = \frac{\pi f_{101} \mu_0 a b d (a^2 + d^2)}{R_s [2b(a^3 + d^3) + ad(a^2 + d^2)]} \quad \left(R_s = \sqrt{\frac{\pi f_{101} \mu_0}{\sigma}} \right)$$

$$= \frac{\sqrt{\pi f_{101} \mu_0 \sigma} a b d (a^2 + d^2)}{2b(a^3 + d^3) + ad(a^2 + d^2)}$$

$$= 6869.$$

From Eqs. (9-114) and (9-115):

$$W_e = \frac{1}{4} \epsilon_0 \mu_0^2 a^3 b d f_{101}^2 H_0^2 = 0.0773 \times 10^{-12} \text{ (J)} = 0.0773 \text{ (pJ)},$$

$$W_m = \frac{\mu_0}{16} a b d \left(\frac{a^2}{d^2} + 1 \right) H_0^2 = 0.0773 \text{ (pJ)} = W_e.$$

P. 9-18 $\epsilon_r = 2.5$.

a) $f_{101} = \frac{u}{2} \sqrt{\frac{1}{a^2} + \frac{1}{d^2}} = \frac{1}{\sqrt{\epsilon_r}} (f_{101})_{\epsilon_0} = 3.037 \text{ (GHz)}.$

b) $Q_{101} = \frac{1}{(\epsilon_r)^{1/4}} (Q_{101})_{\epsilon_0} = 5,462.$

c) $W_e = (W_e)_{\epsilon_0} = 0.0773 \text{ (pJ)} = W_m.$

P. 9-19 $\delta = \frac{1}{\sqrt{\pi f_{101} \mu_0 \sigma}}$, $f_{101} = \frac{c}{2\sqrt{2}a} = \frac{c}{\sqrt{2}a}.$

a) $Q_{101} = \frac{a}{3\delta} = \frac{a}{3} \sqrt{\pi c \mu_0 \sigma / \sqrt{2}a} = 6,500.$

$$19,500(2)^{1/4} = \sqrt{a} \sqrt{\pi 3 \times 10^8 (4\pi 10^{-7}) (1.57 \times 10^7)}$$

$$\longrightarrow a = 0.0289 \text{ (m)} = 2.89 \text{ (cm)}.$$

b) $f_{101} = \frac{c}{\sqrt{2}a} = 7.34 \text{ (GHz)}.$

c) For copper, $\sigma = 5.80 \times 10^7 \text{ (S/m)}.$

$$Q_{101} \propto \sqrt{\sigma} \\ = 6,500 \sqrt{\frac{5.80}{1.57}} = 12,493.$$

P. 9-20 $Q_{101} = \frac{\sqrt{\pi f_{101} \mu_0 \sigma} abd(a^2 + d^2)}{2b(a^3 + d^3) + ad(a^2 + d^2)}.$

a) For $a = d = 1.8b = 0.036 \text{ (m)}$, $b = 0.02 \text{ (m)}.$

$$f_{101} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{d^2}} = 1.179 \times 10^8 \left(\frac{1}{b}\right) = 5.89 \times 10^9 \text{ (Hz)}$$

$$Q_{101} = 10.23 \sqrt{\sigma b} = 11,018.$$

b) For $Q'_{101} = 1.20 Q_{101} \longrightarrow b' = 1.20^2 b = 1.44 \times 0.02 \\ = 0.0288 \text{ (m)} = 2.88 \text{ (cm)}.$

Chapter 10

Antennas and Antenna Arrays

P. 10-1 From Eqs. (10-12) and (10-14):

$$G_D(\theta, \phi) = \frac{4\pi R^2 \mathcal{P}_{av}}{P_r}$$

Maximum G_D at $\mathcal{P}_{av}$ occur at $\theta = \pi/2$.

$$\mathcal{P}_{av} = \frac{DP_r}{4\pi R^2} = \frac{E_0^2}{2\eta_0}$$

$$E_0^2 = \frac{\eta_0 DP_r}{2\pi R^2} ; \quad D = 1.5, \quad P_r = 0.70 \times 15 \times 10^3 \text{ (W)}$$

$$\longrightarrow E_0 = 0.0972 \text{ (V/m)} = 97.2 \text{ (mV/m)}$$

$$H_0 = \frac{E_0}{\eta_0} = 0.258 \text{ (mA/m)}$$

P. 10-2 a) $D = \frac{U_{max}}{U_{av}} \quad U_{max} = 50$

$$\begin{aligned} U_{av} &= \frac{1}{4\pi} \int U d\Omega \\ &= \frac{50}{4\pi} \int_{-\pi/2}^{\pi/2} \int_0^\pi (\sin^2 \theta \cos \phi) \sin \theta d\theta d\phi \\ &= 2.65 \text{ (W/sr)} \end{aligned}$$

$$\longrightarrow D = \frac{50}{2.65} = 18.85, \text{ or } 12.75 \text{ (dB)}$$

b) $U_{av} = \frac{P_r}{4\pi}$

$$\begin{aligned} P_r &= 4\pi U_{av} = 4\pi \times 2.65 = 33.3 \\ &= \frac{1}{2} I_i^2 R_r \end{aligned}$$

$$\longrightarrow R_r = \frac{2P_r}{I_i^2} = \frac{2 \times 33.3}{2^2} = 16.7 \text{ (}\Omega\text{)}$$

P. 10-3 Equation of continuity: $\nabla \cdot \vec{J} = -j\omega\rho$

$$\longrightarrow \rho_e = \frac{j}{\omega} \frac{dI(z)}{dz}$$

a) $I(z) = I_0 \cos 2\pi z \longrightarrow \rho_e = -\frac{I_0}{c} \sin 2\pi z$.

$$\beta = \frac{2\pi}{\lambda} = 2\pi$$

$\longrightarrow$ Wavelength $\lambda = 1$ (m).

b) $I(z) = I_0 (1 - \frac{4}{\lambda}|z|) \longrightarrow \rho_e = \begin{cases} -j \frac{2I_0}{\pi c} & \text{for } z > 0, \\ +j \frac{2I_0}{\pi c} & \text{for } z < 0. \end{cases}$

P. 10-4 $\lambda = \frac{3 \times 10^8}{10^6} = 300$ (m), $\frac{dl}{\lambda} = \frac{15}{300} = \frac{1}{20} \ll 1$ (Hertzian dipole).

a) Radiation resistance, $R_r = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2 = 1.97$ (Ω).

b) Eq. (10-30): $R_s = \sqrt{\frac{\pi f \mu_0}{\sigma}} = \sqrt{\frac{\pi 10^6 (4\pi \times 10^{-7})}{5.80 \times 10^7}} = 2.61 \times 10^{-4}$ (Ω).

Eq. (10-29): $R_L = R_s \left(\frac{dl}{2\pi a}\right) = 0.031$ (Ω) $\longrightarrow \eta_r = \frac{R_r}{R_r + R_L} = 98.5\%$.

c) $\left. \begin{aligned} \text{Eq. (10-14): } P_r &= \frac{I^2 (dl)^2}{12\pi} \eta_0 \beta^2 \\ \text{Eq. (10-10): } |E_\theta|_{\max} &= \left(\frac{I dl}{4\pi R}\right)^2 \frac{\eta_0 \beta^2}{R^2} \end{aligned} \right\} \longrightarrow |E_\theta|_{\max} = \frac{1}{R} \sqrt{90 P_r} = 19$ (mV/m).

P. 10-5 $R_s = \sqrt{\frac{\pi f \mu_0}{\sigma}} = \sqrt{\frac{\pi (10)^8 (4\pi 10^{-7})}{1.57 \times 10^7}} = 5.01 \times 10^{-3}$ (Ω).

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3$$
 (m).

Dipole length = 1.5 (m) $\longrightarrow \frac{\lambda}{2}$ dipole.

$$\longrightarrow R_r = 73.1$$
 (Ω).

$$\begin{aligned} \text{Power lost, } P_L &= \frac{R_s}{2\pi a} \int_{-\lambda/4}^{\lambda/4} \frac{1}{2} (I_0 \cos \beta z)^2 dz \\ &= \frac{R_s}{2\pi a} \left(\frac{I_0^2}{2}\right) \frac{1}{\beta} \int_{-\pi/2}^{\pi/2} \cos^2 x dx = 0.598 \left(\frac{I_0^2}{2}\right). \end{aligned}$$

$$P_r = \left(\frac{I_0^2}{2}\right) R_r = 73.1 \left(\frac{I_0^2}{2}\right).$$

$$\eta_r = \frac{P_r}{P_r + P_L} = \frac{73.1}{73.1 + 0.598} = 0.992, \text{ or } 99.2\%.$$

P. 10-6

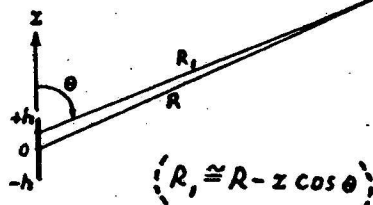
a)
$$E_\theta = j \frac{I_0 \eta_0 \beta \sin \theta}{4\pi R} e^{-j\beta R} \int_{-h}^h \left(1 - \frac{|z|}{h}\right) e^{j\beta z \cos \theta} dz$$

$$= j \frac{I_0 \eta_0 \beta \sin \theta}{2\pi R} e^{-j\beta R} \int_0^h \left(1 - \frac{z}{h}\right) \cos(\beta z \cos \theta) dz$$

$$= \frac{j 60 I_0}{(\beta h) R} e^{-j\beta R} F(\theta),$$

$$H_\phi = \frac{E_\theta}{\eta_0} = \frac{j I_0}{(\beta h) 2\pi R} e^{-j\beta R} F(\theta),$$

$$F(\theta) = \frac{\sin \theta [1 - \cos(\beta h \cos \theta)]}{\cos^2 \theta}.$$



In case $\beta h \ll 1$, $\cos(\beta h \cos \theta) \approx 1 - \frac{1}{2}(\beta h \cos \theta)^2$, and

$$F(\theta) \approx \frac{1}{2}(\beta h)^2 \sin \theta.$$

$$\therefore E_\theta = \frac{j 60 I_0}{R} e^{-j\beta R} \left(\frac{1}{2} \beta h \sin \theta\right) = \frac{j 30 \beta h}{R} I_0 e^{-j\beta R} \sin \theta,$$

$$H_\phi = \frac{j I_0}{2\pi R} e^{-j\beta R} \left(\frac{1}{2} \beta h \sin \theta\right) = \frac{j \beta h}{4\pi R} I_0 e^{-j\beta R} \sin \theta.$$

b)
$$P_r = \frac{1}{2} \int_0^{2\pi} \int_0^\pi E_\theta H_\phi^* R^2 \sin \theta d\theta d\phi = \frac{I_0^2}{2} \left[80 \pi^2 \left(\frac{h}{\lambda}\right)^2 \right],$$

$$R_r = P_r / \left(\frac{1}{2} I_0^2\right) = 20 \pi^2 \left(\frac{2h}{\lambda}\right)^2.$$

c)
$$D = \frac{4\pi |E_{\max}|^2}{\int_0^{2\pi} \int_0^\pi |E_\theta(\theta)|^2 \sin \theta d\theta d\phi} = \frac{2}{\int_0^\pi \sin^3 \theta d\theta} = 1.5 \rightarrow 10 \log_{10} D = 1.76 \text{ (dB)}.$$

P. 10-7

$$f = 180 \times 10^3 \text{ (Hz)} \rightarrow \lambda = \frac{c}{f} = \frac{3 \times 10^8}{180 \times 10^3} = 1.667 \text{ (m)}.$$

$h = 40 \text{ (m)} \ll \lambda$, with triangular current distribution.

a) From Problem P. 10-6, we have, $(\beta = \frac{2\pi}{\lambda})$

$$|E_\theta|_{\max} = \frac{30 \beta h}{R} I_0 = \frac{30 \times 2\pi \times 40}{1667 \times 160} \times 100 = 2.83 \text{ (mV/m)}.$$

$$|H_\phi|_{\max} = \frac{1}{\eta_0} |E_\theta|_{\max} = \frac{2.83 \times 10^{-3}}{377} = 7.51 \times 10^{-6} \text{ (A/m)} = 7.51 \text{ (}\mu\text{A/m)}.$$

b)
$$P_r = \frac{1}{2} \int_0^{\pi/2} \frac{|E_\theta|^2}{\eta_0} 2\pi R^2 \sin \theta d\theta = \frac{30^2}{120} (\beta h I_0)^2 \int_0^{\pi/2} \sin^3 \theta d\theta$$

$$= \frac{30}{4} (\beta h I_0)^2 \left(\frac{2}{3}\right) = 5 (\beta h I_0)^2 = \frac{I_0^2}{2} \left[40 \pi^2 \left(\frac{h}{\lambda}\right)^2 \right]$$

$$= \frac{100^2}{2} \times 40 \pi^2 \left(\frac{40}{1667}\right)^2 = 1,136.5 \text{ (W)} \approx 1.14 \text{ (kW)}.$$

c)
$$R_r = 2 P_r / I_0^2 = 0.227 \text{ (}\Omega\text{)}.$$

P. 10-9 a) E-plane pattern function for Hertzian dipole is, from Eq. (10-10),

$$F_a(\theta) = \sin \theta.$$

$$\text{Max. } F_a(\theta) = 1 \text{ at } \theta_0 = 90^\circ.$$

$$\text{Half-power points at } F_a(\theta_1) = F_a(\theta_2) = \frac{1}{\sqrt{2}}.$$

$$\longrightarrow \theta_1 = 45^\circ, \quad \theta_2 = 135^\circ.$$

$$\longrightarrow \text{Beamwidth } \Delta\theta = \theta_2 - \theta_1 = 90^\circ.$$

b) E-plane pattern function for half-wave dipole is, from Eq. (10-38),

$$F_b(\theta) = \frac{\cos[(\pi/2)\cos\theta]}{\sin\theta}.$$

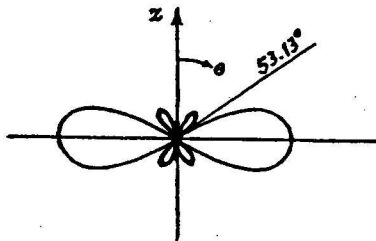
$$\text{Max. } F_b(\theta) = 1 \text{ at } \theta_0 = 90^\circ.$$

$$\text{Half-power points at } F_b(\theta'_1) = F_b(\theta'_2) = \frac{1}{\sqrt{2}}.$$

$$\longrightarrow \text{Beamwidth } \Delta\theta' = \theta'_2 - \theta'_1 = 129^\circ - 51^\circ = 78^\circ.$$

P. 10-10 Use Eq. (10-37):

$$F(\theta) = \frac{\cos(\beta h \cos\theta) - \cos\beta h}{\sin\theta}.$$



$$\text{For } 2h = 1.25\lambda,$$

$$|F(\theta)| = \left| \frac{\cos(1.25\pi \cos\theta) - \cos(1.25\pi)}{\sin\theta} \right|$$

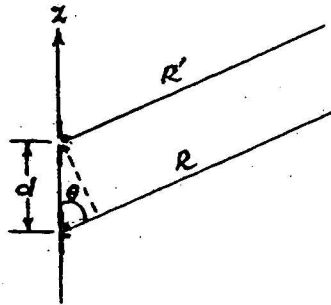
Width of main beam between the first nulls

$$= 2 \times 53.13^\circ = 106.26^\circ.$$

P. 10-11 Use Eq. (10-10) for Hertzian dipoles.

$$E_{\theta} = E_1(\theta) + E_2(\theta) \\ = j \frac{I(2h)}{4\pi} \eta_0 \beta \sin \theta \left(\frac{e^{-j\beta R}}{R} + \frac{e^{-j\beta R'}}{R'} \right)$$

In the far zone, $R' \cong R - d \cos \theta$.

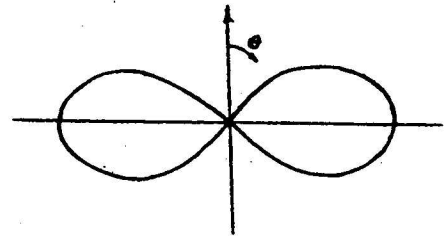


$$a) E_{\theta} \cong j \frac{I(2h)}{4\pi R} \eta_0 \beta \sin \theta \cdot e^{-j\beta R} \\ \cdot (1 + e^{j\beta d \cos \theta}) \\ = j \frac{60 I h}{R} 2 \beta e^{-j\beta (R - \frac{d}{2} \cos \theta)} F(\theta),$$

where $F(\theta) = \sin \theta \cos(\frac{\beta d}{2} \cos \theta)$.

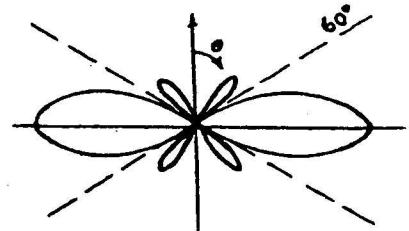
b) $d = \lambda/2$.

$$|F(\theta)| = \left| \sin \theta \cos\left(\frac{\pi}{2} \cos \theta\right) \right|.$$



c) $d = \lambda$,

$$|F(\theta)| = \left| \sin \theta \cos(\pi \cos \theta) \right|.$$



P. 10-12 For an array of identical elements spaced a distance d apart, we have, from Eq. (10-54),

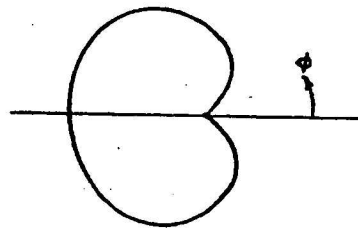
$$|E| = \frac{2E_m}{R_0} |F(\theta, \phi)| \left| \cos \frac{\psi}{2} \right|,$$

$$\text{where } \psi = \beta d \sin \theta \cos \phi + \xi.$$

In the H-plane of a dipole: $\theta = \pi/2$, $F(\frac{\pi}{2}, \phi) = 1$.

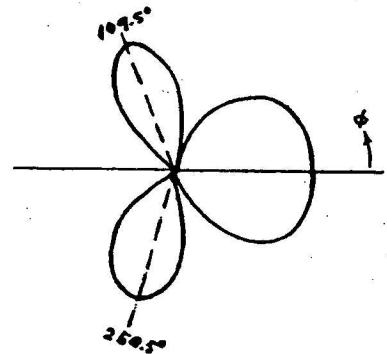
a) $d = \frac{\lambda}{4}$, $\xi = \frac{\pi}{2}$.

$$|A(\phi)| = \left| \cos \frac{\psi}{2} \right| = \left| \cos \left[\frac{\pi}{4} (1 + \cos \phi) \right] \right|.$$



b) $d = \frac{3\lambda}{4}$, $\xi = \frac{\pi}{2}$.

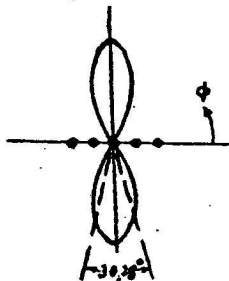
$$|A(\phi)| = \left| \cos \left(\frac{3\pi}{4} \cos \phi + \frac{\pi}{4} \right) \right|.$$



P. 10-13 Five-element broadside binomial array.

a) Relative excitation amplitudes: 1:4:6:4:1.

b) Array factor: $|A(\phi)| = \left| \cos \left(\frac{\pi}{2} \cos \phi \right) \right|^4$.



c) $\cos \left(\frac{\pi}{2} \cos \phi \right) = (\sqrt{2})^{-1/4}$

$$\rightarrow \phi = 74.86^\circ.$$

Half-power beamwidth

$$= 2(90^\circ - 74.86^\circ)$$

$$= 30.28^\circ.$$

For uniform array, from Eq. (11-89):

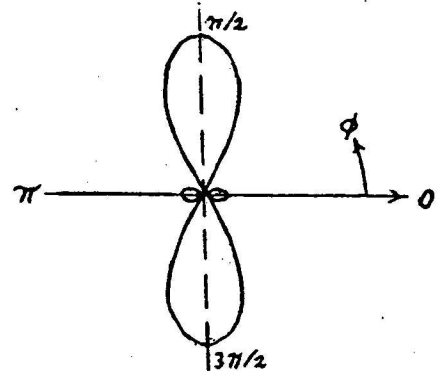
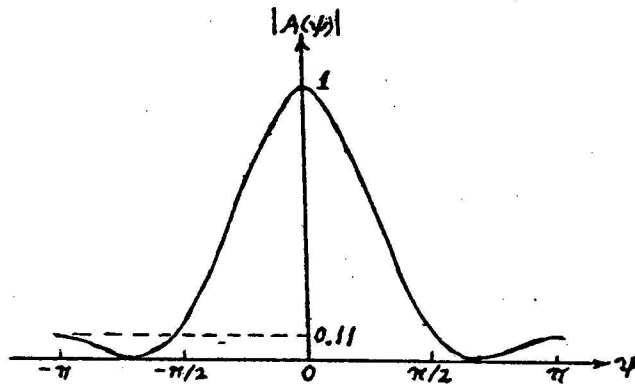
$$\frac{1}{5} \left| \frac{\sin(\frac{5\pi}{2} \cos \phi)}{\sin(\frac{\pi}{2} \cos \phi)} \right| = \frac{1}{\sqrt{2}} \rightarrow \phi = 79.61^\circ$$

Half-power beamwidth for 5-element uniform array
with $\lambda/2$ spacing $= 2(90^\circ - 79.61^\circ) = 20.78^\circ$.

P.10-14 The normalized array factor of the five-element tapered array is

$$\begin{aligned} |A(\psi)| &= \frac{1}{9} \left| 1 + 2e^{j\psi} + 3e^{j2\psi} + 2e^{j3\psi} + e^{j4\psi} \right| \\ &= \frac{1}{9} \left| e^{j2\psi} [3 + 2(e^{j\psi} + e^{-j\psi}) + (e^{j3\psi} + e^{-j3\psi})] \right| \\ &= \frac{1}{9} \left| 3 + 4\cos\psi + 2\cos 2\psi \right| \end{aligned}$$

A graph of $|A(\psi)|$ vs. ψ is shown below on the left.



For broadside operation: $\xi = 0$, $\psi = \beta d \cos \phi = \pi \cos \phi$.

$$|A(\phi)| = \frac{1}{9} \left| 3 + 4\cos(\pi \cos \phi) + 2\cos(2\pi \cos \phi) \right|$$

This is plotted above on the right. The first sidelobe level is 0.11, or $20\log_{10}(1/0.11) = 19.2$ (dB) down from the main-beam radiation. This compares with 0.25, or 12 (dB) down for the five-element uniform broadside array shown in Fig. 10-11.

P.10-15 From Eqs. (10-39) and (10-60):

$$|E_\theta| = \frac{\gamma 60 I_m N_1 N_2}{R} e^{-j\beta R} \left| \frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} A_x(\psi_x) A_y(\psi_y) \right|$$

$$\text{where } A_x(\psi_x) = \frac{1}{N_1} \frac{\sin(N_1 \psi_x / 2)}{\sin(\psi_x / 2)}, \quad \psi_x = \frac{\beta d_1}{2} \sin \theta \cos \phi;$$

$$A_y(\psi_y) = \frac{1}{N_2} \frac{\sin(N_2 \psi_y / 2)}{\sin(\psi_y / 2)}, \quad \psi_y = \frac{\beta d_2}{2} \sin \theta \cos \phi.$$

$$|F(\theta, \phi)| = \frac{1}{N_1 N_2} \left| \left[\frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} \right] \frac{\sin(\frac{N_1 \psi_x}{2}) \sin(\frac{N_2 \psi_y}{2})}{\sin(\frac{\psi_x}{2}) \sin(\frac{\psi_y}{2})} \right|$$

P. 10-16 $\ell_e = \frac{1}{I(0)} \int_{-h}^h I(z) dz.$

a) Hertzian dipole of length dl .

$$I(z) = I(0), \quad h = \frac{1}{2} dl, \quad \sin(\beta \frac{dl}{2}) \approx \beta \frac{dl}{2}.$$

$$\ell_e = \int_{-dl/2}^{dl/2} \cos \beta z dz = dl.$$

b) Half-wave dipole with $h = \lambda/4$ and $I(z) = I(0) \cos \beta z$.

$$\ell_e = \int_{-\lambda/4}^{\lambda/4} \cos \beta z dz = \frac{2}{\beta} \sin(\beta \frac{\lambda}{4}) = \frac{2}{\beta} = \frac{\lambda}{\pi}.$$

c) Half-wave dipole with $h = \lambda/4$ and $I(z) = I(0)(1 - 4|z|/\lambda)$.

$$\ell_e = \int_{-\lambda/4}^{\lambda/4} (1 - 4|z|/\lambda) dz = 2 \int_0^{\lambda/4} (1 - 4z/\lambda) dz = \frac{\lambda}{4}.$$

P. 10-17 $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{300 \times 10^6} = 1 \text{ (m)}.$

Half-wave dipole with sinusoidal current distribution

$$I(z) = I(0) \sin \beta (\frac{\lambda}{4} - |z|) \quad (\beta \frac{\lambda}{4} = \frac{\pi}{2}) \\ = I(0) \cos \beta z.$$

From Eq. (10-85) and problem P. 10-16 (b), $\ell_e = \frac{\lambda}{\pi}.$

From Eq. (10-35), we have, for $\theta = \pi/2$,

$$|E_i| = \frac{I(0) \eta_0 \beta}{4 \pi R} \ell_e = \frac{60}{\lambda R} I(0).$$

$$P_r = \frac{1}{2} I^2(0) R_r \longrightarrow I(0) = \sqrt{\frac{2 P_r}{R_r}} = \sqrt{\frac{2 \times 2000}{73.1}} = 7.40 \text{ (A)}.$$

$$|E_i| = \frac{60 \times 7.40}{1 \times 150} = 2.96 \text{ (V/m)}.$$

a) $|V_{oc}| = |E_i \ell_e| = 2.96 \times \frac{1}{\pi} = 0.942 \text{ (V)}.$

b) For matched load,

$$P_L = \frac{V_{oc}^2}{8 R_r} = \frac{0.942^2}{8 \times 73.1} = 1.52 \times 10^{-3} \text{ (W)} = 1.52 \text{ (mW)}.$$

P. 10-18 Eq. (10-80): $P_L = \frac{D_1 D_2 \lambda^2}{(4\pi r)^2} P_t$.

$$r = 150 \text{ (m)}, P_t = 2 \times 10^3 \text{ (W)}, \lambda = \frac{c}{f} = \frac{3 \times 10^8}{300 \times 10^6} = 1 \text{ (m)}.$$

a) Parallel half-wave dipoles: $D_1 = D_2 = 1.64$.

$$P_L = \frac{1.64 \times 1.64 \times 1^2}{(4\pi \times 150)^2} \times 2 \times 10^3 = 1.514 \times 10^{-3} \text{ (W)} \\ = 1.514 \text{ (mW)}.$$

b) Parallel Hertzian dipoles: $D_1 = D_2 = 1.50$.

$$P_L = 1.514 \times \left(\frac{1.50}{1.64} \right)^2 = 1.267 \text{ (mW)}.$$

P. 10-19 From Eqs. (10-12) and (10-14):

$$G_D = \frac{4\pi U(\theta, \phi)}{P_r} = \frac{4\pi R^2 \mathcal{P}_{av}}{P_r} \quad (1)$$

Using Eqs. (10-40) and (10-42) in (1):

$$G_D(\theta) = 1.64 \left[\frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} \right]^2 \quad (2)$$

a) Substituting (2) in Eq. (10-75):

$$A_e(\theta) = \frac{\lambda^2}{4\pi} G_D(\theta) = 0.13 \lambda^2 \left[\frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} \right]^2.$$

b) Max. value of $A_e(\theta)$ for $f = 10^8 \text{ (Hz)}$, $\lambda = \frac{c}{f} = 3 \text{ (m)}$
occurs at $\theta = \frac{\pi}{2}$. $A_e(\frac{\pi}{2}) = 0.13 \lambda^2 = 1.17 \text{ (m}^2\text{)}.$

c) Max. value of $A_e(\theta)$ for $f = 2 \times 10^8 \text{ (Hz)}$, $\lambda = 1.5 \text{ (m)}$:

$$A_e(\frac{\pi}{2}) = 0.13 \times 1.5^2 = 0.29 \text{ (m}^2\text{)},$$

which is smaller than $A_e(\frac{\pi}{2})$ for $f = 10^8 \text{ (Hz)}$.

because the wavelength is shorter at $f = 2 \times 10^8 \text{ (Hz)}$.

P. 10-20 Antenna gain: $10 \log_{10} G_D = 20 \text{ (dB)}$

$$\rightarrow G_D = 100.$$

$$f = 3 \times 10^9 \text{ (Hz)} \rightarrow \lambda = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^9} = 0.1 \text{ (m)}.$$

a) Power density at target, $\mathcal{P}_T = \frac{P_t}{4\pi r^2} G_D$.

$$\mathcal{P}_T = \frac{E_T^2}{2\eta_0} = \frac{120 \times 10^3 \times 100}{4\pi(8 \times 10^3)^2} = 0.0149 \text{ (W/m}^2\text{)}$$

$$\rightarrow E_T = \sqrt{0.0149 \times 2 \times 377} = 3.35 \text{ (V/m)}.$$

b) Power intercepted by target $= \sigma_{bs} \mathcal{P}_T = 15 \times 0.0149 = 0.224 \text{ (W)}.$

c) Scattered power density at radar antenna

$$\mathcal{P}_s = \frac{\sigma_{bs} \mathcal{P}_T}{4\pi r^2} = \frac{0.224}{4\pi(8 \times 10^3)^2} = 2.78 \times 10^{-10} \text{ (W)}$$

$$\begin{aligned} \text{Reflected power absorbed by antenna} &= \mathcal{P}_s A_e \\ &= 2.78 \times 10^{-10} \left(\frac{\lambda^2}{4\pi} G_D \right) = 2.78 \times 10^{-10} \left(\frac{0.1^2}{4\pi} \times 100 \right) = 22.1 \times 10^{-12} \text{ (W)} \\ &= 22.1 \text{ (pW)}. \end{aligned}$$

P. 10-21 Earth radius = 6,380 (km).

Altitude of geosynchronous satellites = 36,500 (km)

$$\begin{aligned} \text{Geosynchronous orbit radius} &= 6,380 + 36,500 \\ &= 42,880 \text{ (km)}. \end{aligned}$$

$$\psi = \sin^{-1} \left(\frac{6,380}{42,880} \right) = 8.56^\circ$$

$$\psi' = 90^\circ - 8.56^\circ = 81.44^\circ$$

a) Two satellites cover only $2 \times (2\psi') = 326^\circ < 360^\circ$.

Use three satellites in equatorial plane: $3 \times (2\psi') = 489^\circ > 360^\circ$.

Polar regions are not covered because $\psi' < 90^\circ$.

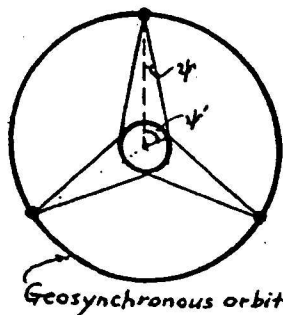
b) Let P_t = Power transmitted by satellite antenna.

$$\mathcal{P}_c = \text{Power density within the cone} = \frac{G_D}{4\pi r^2} P_t.$$

$$\text{Area of cone cap on earth} = \int_0^{2\pi} \int_0^\psi r^2 \sin \theta \, d\theta \, d\phi$$

$$= 2\pi r^2 (1 - \cos \psi) \approx 2\pi r^2 (\psi^2/2) = \pi (r\psi)^2.$$

$$\therefore P_t = \pi (r\psi)^2 \mathcal{P}_c \rightarrow \psi = \frac{1}{r} \sqrt{P_t / \pi \mathcal{P}_c} = 2 / \sqrt{G_D} \rightarrow \text{Main-lobe beamwidth} = 2\psi = \frac{4}{\sqrt{G_D}}.$$



P. 10-22 a) From Eq. (10-80):

$$P_t = \frac{(4\pi r)^2}{G_e G_s \lambda_s^2} P_L,$$

where the subscripts e and s denote earth and Satellite respectively.

$$\lambda_e = \frac{3 \times 10^8}{14 \times 10^9} = 2.14 \times 10^{-2} \text{ (m)},$$

$$G_e = 10^{(55/10)} = 3.16 \times 10^5;$$

$$\lambda_s = \frac{3 \times 10^8}{12 \times 10^9} = 2.50 \times 10^{-2} \text{ (m)},$$

$$G_s = 10^{(35/10)} = 3.16 \times 10^3.$$

$$r = 3.65 \times 10^7 \text{ (m)}, \quad P_L = 8 \times 10^{-12} \text{ (W)}.$$

$$\longrightarrow P_t = 2.7 \text{ (W)}.$$

b) From Eq. (10-84): $P_t = \frac{4\pi}{\sigma_{bs}} \left(\frac{\lambda_e r^2}{A_e} \right)^2 P_L$

$$A_e = \frac{\lambda_e^2}{4\pi} G_e = \frac{(2.14 \times 10^{-2})^2}{4\pi} \times 3.16 \times 10^5$$
$$= 11.5 \text{ (m}^2\text{)}.$$

$$P_t = \frac{4\pi}{25} \left(\frac{2.14 \times 10^{-2} \times 3.65^2 \times 10^{14}}{11.5} \right)^2 \times 0.5 \times 10^{-12}$$
$$= 1.54 \times 10^{12} \text{ (W)} = 1.54 \text{ (TW)}.$$